

Candidate Name

Centre Number

Candidate Number



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

CHEMISTRY

PAPER 2 Theory

4024/2

2 hours

JUNE 2026 SESSION

Additional materials:
Electronic calculator
Answer booklet

0798

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page and on all separate answer paper used.

Answer **all** questions in **Section A** and any **four** from **Section B**.

At the end of the examination, fasten any separate answer paper used securely to the question paper.

Enter the numbers of **Section B** questions you have answered in the grid.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

A copy of the Periodic Table is printed on page 16.

FOR EXAMINER'S USE**Section A****Section B****TOTAL**

This question paper consists of 16 printed pages.

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[Turn over

SECTION A

Answer **all** questions from this section.

Write your answers on the spaces provided in the question paper.

1. (a) **Fig 1.1** shows the positions of elements **X**, **Y** and **Z** in a periodic table

	X							Y									Z

Fig 1.1

- (i) Identify the letter that represents an element that

1. exists as a di-atomic molecule:

2. is a transition element

[2]

- (ii) State the type of bonding in the element represented by **Y**.

[1]

- (b) Phosphorous forms an ion $^{31}_{15}\text{P}^{3-}$.

- (i) Deduce the number of electrons, protons and neutrons in the ion

1. *electrons*,

2. *protons*,

3. *neutrons*.

[3]

- (ii) State the
1. formula of the oxide of phosphorous,
 2. acid base nature of the oxide.

[2]

2. (a) Fig 2.1 shows a set up of an experiment in a school laboratory.

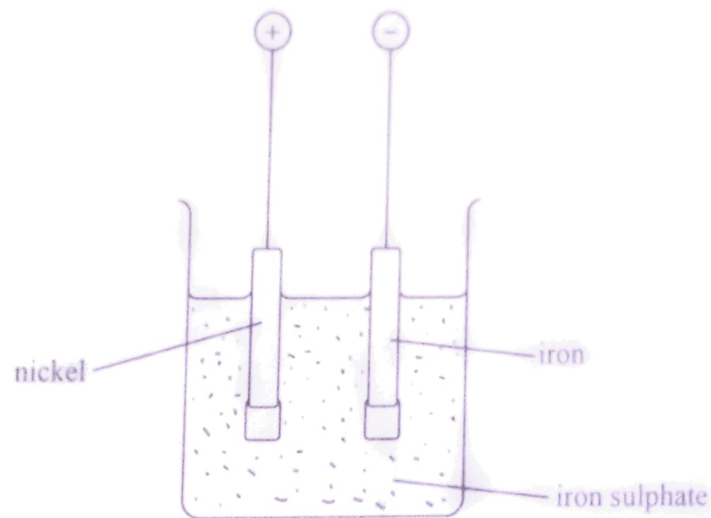


Fig 2.1

- (i) Suggest why there were **no** observable changes that occurred after sometime.

[1]

- (ii) Describe and explain how the error in the set up can be corrected.

[2]

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(b) (i) Give any **two** reasons for coating metals.

1

2

[2]

(ii) State any **three** factors that will affect the quality of an electroplating process.

1

2

3

[3]

- 3 (a) Fig 3.1 shows a set up to investigate the reaction between calcium carbonate and 0.1 mol dm^{-3} hydrochloric acid. The readings taken from an electronic balance are shown in Table 3.1. The experiment was repeated using 0.50 mol dm^{-3} hydrochloric acid.

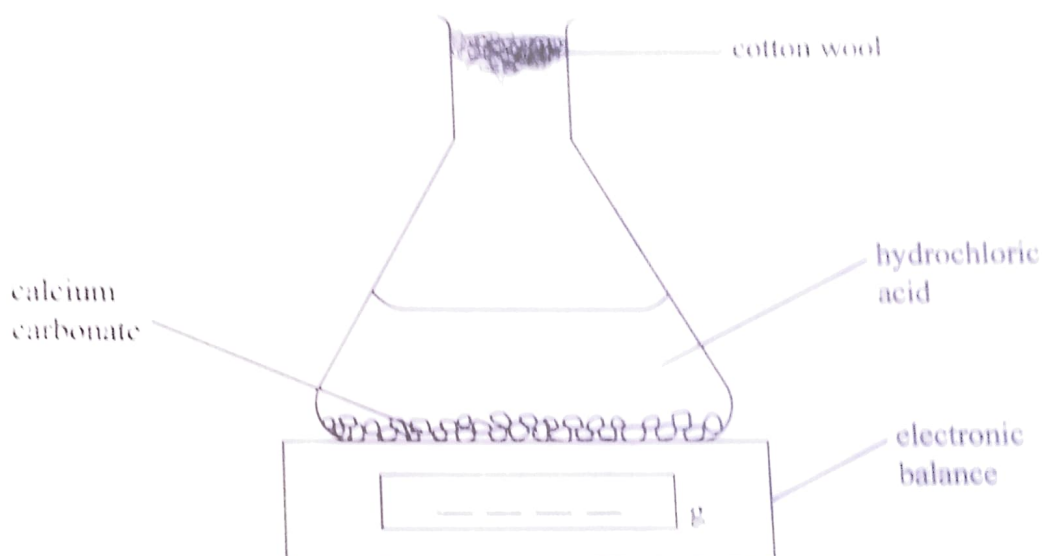


Fig 3.1

Table 3.1

experiment number	initial electronic balance reading	electronic balance reading after 2 minutes
1	118.9	114.4
2	109.4	99.3

- (i) Write a balanced chemical equation for the reaction.

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(ii) Explain the

1. decrease in mass in the two experiments.

.....

.....

[1]

2. difference in the decrease in mass after 2 minutes for the two experiments.

.....

.....

.....

.....

[3]

(b) A mass of 14.2 g of chlorine was reacted with hydrogen to produce hydrogen chloride.

Calculate the enthalpy change for the reaction, given that the molar enthalpy change for the formation of hydrogen chloride is -184 kJ/mol .

enthalpy change for the reaction =

[2]

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4. (a) (i) Define the term *isomer*.

[1]

- (ii) Draw all the possible isomers of butane.

[2]

- (b) (i) Identify the main gas in wood gas which can also be used as a fuel.

[1]

- (ii) State the **two** conditions necessary for the destructive distillation of wood.

1.

2.

[1]

- (iii) Name the type of wood that would give the best quality of charcoal

[1]

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(iv) Give any **two** advantages of using charcoal as a fuel

1.

2.

[2]

5. (a)

Table 5.1 is an incomplete table showing the advantages and the disadvantages of using herbs for medicinal purposes.

Complete **Table 5.1** by stating any **three** advantages and any **three** disadvantages of using herbs for medicinal purposes.

Table 5.1

advantages	disadvantages

[6]

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(b) Describe any **one** use of herbs in

1. *the food industry.*

2. *agriculture.*

[2]

SECTION B

Answer any **four** questions from this section.

Write your answer on the separate paper provided.

6. (a) (i) Draw a dot and cross diagram to show the electronic structure of ammonia. [1]
- (ii) Ammonia can be produced industrially as shown in Fig 6.1.

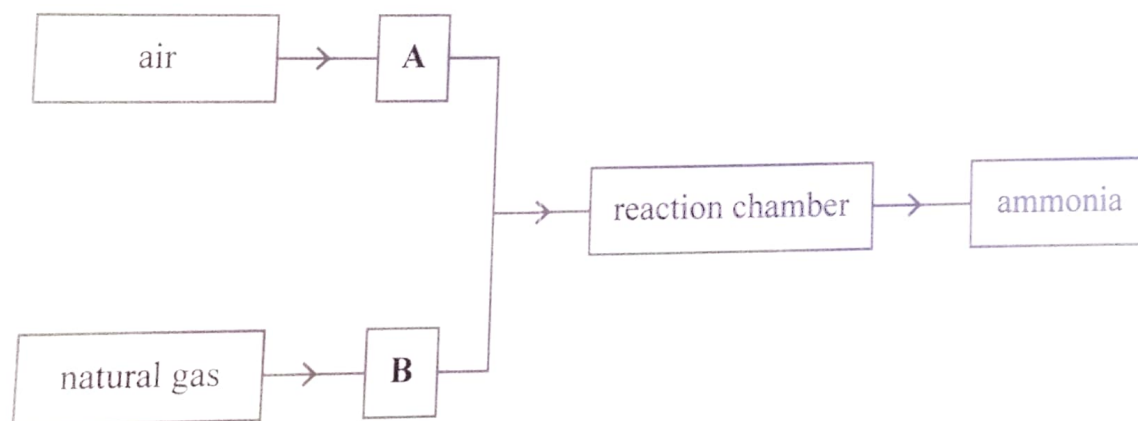


Fig 6.1

Identify substance

1. A
 2. B
- [2]

- (iii) State the **two** conditions required in the reaction chamber for a high yield of ammonia to be obtained. [3]

- (b) A chemistry student prepared 500.00 cm^3 of 1.00 mol/dm^3 of ammonium dichromate, $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$.

Calculate the mass of ammonium dichromate used to prepare this reagent. [2]

- (c) (i) Fig 6.2 shows a label on a container of chemical waste.

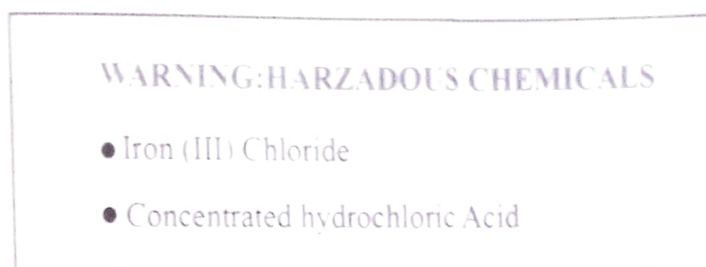


Fig 6.2

Explain why the chemical waste is hazardous.

[2]

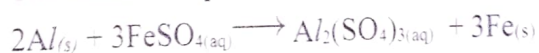
- (ii) Describe the most suitable methods of disposing the chemical waste.

[2]

- (iii) Suggest the environmental effects of improper disposal of chemicals.

[3]

7. (a) A mass of 0.27 g of aluminium were reacted with excess iron (II) sulphate. The chemical equation for the reaction is shown.



- (i) Name the type of reaction.

[1]

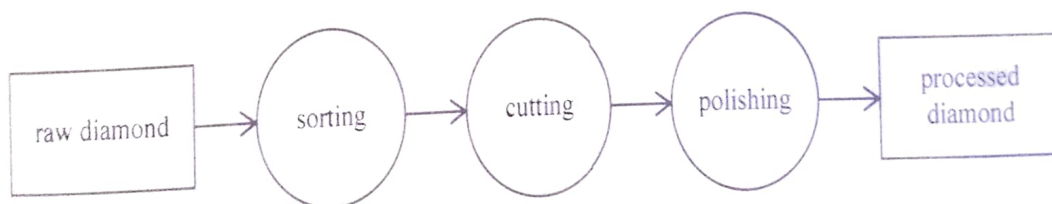
- (ii) Calculate the number of moles of Al that reacted.

[1]

- (iii) Calculate the mass of $\text{Al}_2(\text{SO}_4)_3$ formed.

[3]

- (b) The flow chart shows some stages involved in the processing of diamonds.



- (i) Describe what happens during:

1. sorting,
2. cutting,
3. polishing.

[3]

(ii) Suggest any **two** benefits of a processed diamond. [2]

(c) Bronze and brass are two alloys of copper.

(i) Define the term *alloy*. [1]

(ii) State the composition of:

1 bronze,

2 brass.

[2]

(iii) State any two advantages of using alloys. [2]

8. (a) **Fig 8.1** shows a flow chart of how iron is produced in a blast furnace.

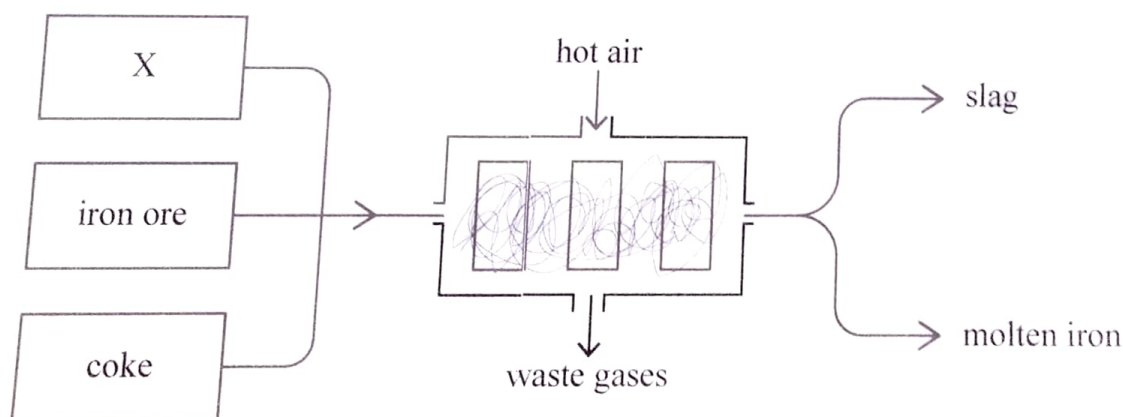


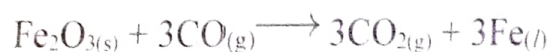
Fig 8.1

(i) Identify substance X. [1]

(ii) State and explain the function of X. [2]

(iii) Name one of the gases present in the waste gases. [1]

(iv) Fe_2O_3 is reduced to Fe according to the following equation:



Calculate the mass of Iron produced from 3.20 g of the iron ore. [3]

- (b) **Table 8.1** shows some physical properties of some organic compounds

Table 8.1

organic compound	boiling point	solubility in water
hexane	68.7	insoluble
octane	125.6	insoluble
ethanol	78.37	soluble
carboxylic acid	118	soluble

Suggest, with a reason the most suitable procedure of separating a mixture of.

(i) hexane, octane and ethanol.

(ii) carboxylic acid and hexane.

[5]

- (c) The cells shown in **Table 8.2** were connected to a bulb.

Table 8.2

cell	electrodes used
A	Mg/Cu
B	Fe/Cu
C	Zn/Cu

- (i) Identify, with a reason, the cell that will give the brightest light. [2]

- (ii) Suggest a suitable electrolyte that can be used in the cells. [1]

9. (a) The most common waste disposal methods in a school are incineration, landfilling and recycling.

State any **one** advantage and any **one** disadvantage for each of these waste disposal methods. [6]

- (b) (i) A sample of water was purified in a school laboratory.
Name the methods that can be used to purify the water sample. [2]

- (ii) Explain how each of the methods in **b(i)** improves the purity. [2]

- (c) (i) Exhaust fumes contain many pollutants.
Name any **two** pollutants from exhaust fumes. [2]

- (ii) State any **two** effects of the named pollutants in **(i)** to the environment. [2]

- (iii) Suggest any **one** way of minimising air pollution by exhaust gases. [1]

10. (a) (i) Draw the structural formula for propene. [1]

- (ii) Name the type of reaction undergone when propene reacts with hydrogen. [1]

- (iii) Describe how bromine water can be used to distinguish propene from propane. [2]

- (b) The empirical formula of an organic compound, **R** is CH_2O .

- (i) Deduce the molecular formula of **R**, given that the relative molecular mass is 60. [3]

- (ii) Draw the structural formula of **R**. [1]

- (iii) State the homologous series to which **R** belongs. [1]

- (c) **Fig 10.1** shows one of the uses of some organic compounds.

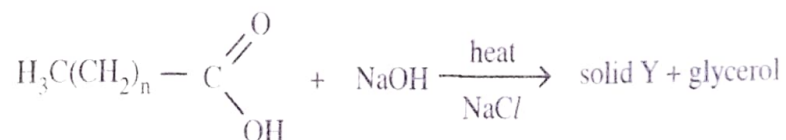


Fig 10.1

- (i) Identify solid Y. [1]
- (ii) Name the process shown in **Fig 10.1**. [1]
- (iii) Describe a simple chemical test for solid Y. [2]
- (iv) Suggest any **two** methods of adding value to solid Y. [2]

DATA SHEET

The Periodic Table of the Elements



18559

Group		I	II	III	IV	V	VI	VII	O
		1 Hydrogen H							2 Helium He
				11 Boron B	12 Carbon C	14 Nitrogen N	16 Oxygen O	17 Fluorine F	18 Neon Ne
				27 Aluminum Al	28 Silicon Si	31 Phosphorus P	32 Sulfur S	35.5 Chlorine Cl	40 Argon Ar
				49 Copper Cu	50 Zinc Zn	55 Cadmium Cd	56 Mercury Hg	80 Thallium Tl	81 Lead Pb
				75 Antimony Sb	76 Tellurium Te	81 Iodine I	83 Xenon Xe	85 Radium Ra	86 Actinium Ac
				101 Ruthenium Ru	102 Rhodium Rh	106 Palladium Pd	108 Silver Ag	112 Copper Cu	113 Zinc Zn
				135 Cesium Cs	137 Barium Ba	140 Lanthanum La	141 Cerium Ce	143 Praseodymium Pr	144 Neodymium Nd
				167 Francium Fr	169 Radium Ra	171 Actinium Ac	173 Thorium Th	175 Protactinium Pa	177 Uranium U
				223 Francium Fr	225 Radium Ra	227 Actinium Ac	229 Thorium Th	231 Protactinium Pa	233 Uranium U
				287 Tennessine Ts	289 Oganesson Og	291 Livermorium Lv	293 Tennessine Ts	295 Oganesson Og	297 Livermorium Lv

The volume of one mole of any gas is 28 dm³ at room temperature and pressure (r.t.p.)

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